

Coding Challenge 5_Data Wrangling

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1 Question 1 - Downloading and reading two csv files

```
getwd()
```

```
## [1] "C:/Users/admin/Desktop/Assignments-repository"
```

```
library(tidyverse)
```

```
## Warning: package 'tidyverse' was built under R version 4.5.2
```

```
## Warning: package 'ggplot2' was built under R version 4.5.2
```

```
## Warning: package 'tibble' was built under R version 4.5.2
```

```
## Warning: package 'tidyr' was built under R version 4.5.2
```

```
## Warning: package 'readr' was built under R version 4.5.2

## Warning: package 'purrr' was built under R version 4.5.2

## Warning: package 'dplyr' was built under R version 4.5.2

## Warning: package 'stringr' was built under R version 4.5.2

## Warning: package 'forcats' was built under R version 4.5.2

## Warning: package 'lubridate' was built under R version 4.5.2

## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.1.4      v readr      2.1.6
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2    4.0.2      v tibble    3.3.1
## v lubridate  1.9.4      v tidyr     1.3.2
## v purrr      1.2.1
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
diversity<- read.csv("DiversityData.csv")
str(diversity)
```

```
## 'data.frame':    70 obs. of  5 variables:
## $ Code          : chr  "S01_13" "S02_16" "S03_19" "S04_22" ...
## $ shannon       : num  6.62 6.61 6.66 6.66 6.61 ...
## $ invsimpson    : num  211 207 213 205 200 ...
## $ simpson       : num  0.995 0.995 0.995 0.995 0.995 ...
## $ richness      : int  3319 3079 3935 3922 3196 3481 3250 3170 3657 3177 ...
```

```
metadata<- read.csv("Metadata.csv")
str(metadata)
```

```
## 'data.frame':    70 obs. of  5 variables:
## $ Code          : chr  "S01_13" "S02_16" "S03_19" "S04_22" ...
## $ Crop           : chr  "Soil" "Soil" "Soil" "Soil" ...
## $ Time_Point     : int   0 0 0 0 0 0 6 6 6 6 ...
## $ Replicate      : int   1 2 3 4 5 6 1 2 3 4 ...
## $ Water_Imbibed  : chr  "na" "na" "na" "na" ...
```

2 Question 2 - Joining the two dataframes together and creating a new dataframe named alpha

```
library(dplyr)
alpha<-left_join(diversity, metadata, by="Code")
alpha
```

##	Code	shannon	invsimpson	simpson	richness	Crop	Time_Point	Replicate
## 1	S01_13	6.624921	210.72795	0.9952545	3319	Soil	0	1
## 2	S02_16	6.612413	206.86664	0.9951660	3079	Soil	0	2
## 3	S03_19	6.660853	213.01843	0.9953056	3935	Soil	0	3
## 4	S04_22	6.660671	204.69080	0.9951146	3922	Soil	0	4
## 5	S05_25	6.610965	200.25523	0.9950064	3196	Soil	0	5
## 6	S06_28	6.650812	199.32110	0.9949830	3481	Soil	0	6
## 7	S61_32	6.570679	200.23177	0.9950058	3250	Soil	6	1
## 8	S62_35	6.492227	171.27965	0.9941616	3170	Soil	6	2
## 9	S63_38	6.610986	192.08535	0.9947940	3657	Soil	6	3
## 10	S64_41	6.472259	163.99814	0.9939024	3177	Soil	6	4
## 11	S65_44	6.508824	181.69248	0.9944962	2985	Soil	6	5
## 12	S66_47	6.482495	176.90684	0.9943473	2770	Soil	6	6
## 13	S121_51	6.276073	126.56259	0.9920988	3040	Soil	12	1
## 14	S122_54	6.461118	152.98152	0.9934633	3192	Soil	12	2
## 15	S123_57	6.334648	138.92556	0.9928019	2673	Soil	12	3
## 16	S124_60	6.461988	171.13732	0.9941567	3180	Soil	12	4
## 17	S125_63	6.501973	172.97532	0.9942188	3320	Soil	12	5
## 18	S126_66	6.354387	142.61016	0.9929879	2773	Soil	12	6
## 19	S181_70	6.299381	142.64506	0.9929896	2806	Soil	18	1
## 20	S182_74	6.340644	145.48656	0.9931265	3047	Soil	18	2
## 21	S183_78	6.282807	150.39829	0.9933510	2190	Soil	18	3
## 22	S184_82	6.268316	141.14138	0.9929149	2488	Soil	18	4
## 23	S186_90	6.289000	140.45260	0.9928802	2684	Soil	18	6
## 24	C01_11	6.618126	220.66218	0.9954682	3076	Cotton	0	1
## 25	C02_14	6.627206	211.03921	0.9952615	3180	Cotton	0	2
## 26	C03_17	6.616958	216.06631	0.9953718	2938	Cotton	0	3
## 27	C04_20	6.626465	215.93901	0.9953691	3371	Cotton	0	4
## 28	C05_23	6.642822	211.08960	0.9952627	3435	Cotton	0	5
## 29	C06_26	6.679131	216.31351	0.9953771	3629	Cotton	0	6
## 30	C61_30	6.454741	170.03639	0.9941189	2767	Cotton	6	1
## 31	C62_33	6.484032	172.35279	0.9941979	3377	Cotton	6	2
## 32	C63_36	6.517958	173.41489	0.9942335	3804	Cotton	6	3
## 33	C64_39	6.476069	167.13138	0.9940167	3204	Cotton	6	4
## 34	C65_42	6.569722	197.01186	0.9949242	3250	Cotton	6	5
## 35	C66_45	6.482145	172.96394	0.9942184	3009	Cotton	6	6
## 36	C121_49	5.944568	71.55607	0.9860249	2779	Cotton	12	1
## 37	C122_52	6.187755	96.43939	0.9896308	3193	Cotton	12	2
## 38	C123_55	6.129460	81.26646	0.9876948	2859	Cotton	12	3
## 39	C124_58	6.028523	75.49726	0.9867545	2950	Cotton	12	4
## 40	C125_61	6.148179	98.94468	0.9898933	3018	Cotton	12	5
## 41	C126_64	6.347332	150.05708	0.9933359	2946	Cotton	12	6
## 42	C181_68	6.301392	132.36230	0.9924450	3266	Cotton	18	1
## 43	C182_72	6.000205	83.90929	0.9880824	2969	Cotton	18	2
## 44	C183_76	5.981284	82.44127	0.9878702	2636	Cotton	18	3
## 45	C184_80	5.578566	50.73174	0.9802885	2043	Cotton	18	4
## 46	C185_84	6.064655	87.82732	0.9886140	3113	Cotton	18	5
## 47	SB01_12	6.644864	216.86110	0.9953888	3203	Soybean	0	1
## 48	SB02_15	6.615662	211.32573	0.9952680	3055	Soybean	0	2

## 49	SB03_18	6.693987	230.45439	0.9956607	3595 Soybean	0	3
## 50	SB04_21	6.647502	234.80343	0.9957411	3253 Soybean	0	4
## 51	SB05_24	6.605749	198.57265	0.9949641	3187 Soybean	0	5
## 52	SB06_27	6.640696	215.26494	0.9953546	3190 Soybean	0	6
## 53	SB61_31	6.044229	89.13912	0.9887816	2371 Soybean	6	1
## 54	SB62_34	6.437589	154.21624	0.9935156	3248 Soybean	6	2
## 55	SB63_37	6.194632	83.11681	0.9879687	2976 Soybean	6	3
## 56	SB64_40	6.117393	87.20257	0.9885324	3006 Soybean	6	4
## 57	SB65_43	5.439798	29.48338	0.9660826	2809 Soybean	6	5
## 58	SB66_46	6.195816	108.22394	0.9907599	2680 Soybean	6	6
## 59	SB121_50	4.393341	12.39587	0.9193280	2508 Soybean	12	1
## 60	SB122_53	5.630929	52.97931	0.9811247	2403 Soybean	12	2
## 61	SB123_56	5.579523	48.59842	0.9794232	2752 Soybean	12	3
## 62	SB124_59	5.406651	34.08685	0.9706632	2946 Soybean	12	4
## 63	SB125_62	5.863941	63.33020	0.9842097	3165 Soybean	12	5
## 64	SB126_65	5.738025	57.88780	0.9827252	2705 Soybean	12	6
## 65	SB181_69	5.671024	57.37726	0.9825715	2642 Soybean	18	1
## 66	SB182_73	5.489406	43.16854	0.9768350	2773 Soybean	18	2
## 67	SB183_77	5.713960	60.47882	0.9834653	2454 Soybean	18	3
## 68	SB184_81	5.467076	44.06798	0.9773078	2365 Soybean	18	4
## 69	SB185_85	5.729473	55.95864	0.9821297	2789 Soybean	18	5
## 70	SB186_89	5.556356	54.34527	0.9815991	2050 Soybean	18	6
##	Water_Imbibed						
## 1		na					
## 2		na					
## 3		na					
## 4		na					
## 5		na					
## 6		na					
## 7		na					
## 8		na					
## 9		na					
## 10		na					
## 11		na					
## 12		na					
## 13		na					
## 14		na					
## 15		na					
## 16		na					
## 17		na					
## 18		na					
## 19		na					
## 20		na					
## 21		na					
## 22		na					
## 23		na					
## 24		0.0042					
## 25		0.0091					
## 26		0.0013					
## 27		0.0087					
## 28		0.0075					
## 29		0.0046					
## 30		0.058					
## 31		0.044					

```
## 32      0.0569
## 33      0.0841
## 34      0.0535
## 35      0.0029
## 36      0.0651
## 37      0.0527
## 38      0.0675
## 39      0.0545
## 40      0.0623
## 41      0.0021
## 42      0.0034
## 43      0.0632
## 44      0.0514
## 45      0.0577
## 46      0.0554
## 47      0.1664
## 48      0.0942
## 49      0.1248
## 50      0.115
## 51      0.0993
## 52      0.1005
## 53      0.2308
## 54      0.2603
## 55      0.2111
## 56      0.2808
## 57      0.2712
## 58      0.2887
## 59      0.2822
## 60      0.2557
## 61      0.2982
## 62      0.2489
## 63      0.2573
## 64      0.2285
## 65      0.2528
## 66      0.2706
## 67      0.3196
## 68      0.2437
## 69      0.2461
## 70      0.301
```

3 Question 3 - Calculate Pielou's evenness index

```
alpha_even <- alpha %>%
  mutate(even = shannon / log(richness))
alpha_even
```

##	Code	shannon	invsimpson	simpson	richness	Crop	Time_Point	Replicate
## 1	S01_13	6.624921	210.72795	0.9952545	3319	Soil	0	1
## 2	S02_16	6.612413	206.86664	0.9951660	3079	Soil	0	2
## 3	S03_19	6.660853	213.01843	0.9953056	3935	Soil	0	3
## 4	S04_22	6.660671	204.69080	0.9951146	3922	Soil	0	4

## 5	S05_25	6.610965	200.25523	0.9950064	3196	Soil	0	5
## 6	S06_28	6.650812	199.32110	0.9949830	3481	Soil	0	6
## 7	S61_32	6.570679	200.23177	0.9950058	3250	Soil	6	1
## 8	S62_35	6.492227	171.27965	0.9941616	3170	Soil	6	2
## 9	S63_38	6.610986	192.08535	0.9947940	3657	Soil	6	3
## 10	S64_41	6.472259	163.99814	0.9939024	3177	Soil	6	4
## 11	S65_44	6.508824	181.69248	0.9944962	2985	Soil	6	5
## 12	S66_47	6.482495	176.90684	0.9943473	2770	Soil	6	6
## 13	S121_51	6.276073	126.56259	0.9920988	3040	Soil	12	1
## 14	S122_54	6.461118	152.98152	0.9934633	3192	Soil	12	2
## 15	S123_57	6.334648	138.92556	0.9928019	2673	Soil	12	3
## 16	S124_60	6.461988	171.13732	0.9941567	3180	Soil	12	4
## 17	S125_63	6.501973	172.97532	0.9942188	3320	Soil	12	5
## 18	S126_66	6.354387	142.61016	0.9929879	2773	Soil	12	6
## 19	S181_70	6.299381	142.64506	0.9929896	2806	Soil	18	1
## 20	S182_74	6.340644	145.48656	0.9931265	3047	Soil	18	2
## 21	S183_78	6.282807	150.39829	0.9933510	2190	Soil	18	3
## 22	S184_82	6.268316	141.14138	0.9929149	2488	Soil	18	4
## 23	S186_90	6.289000	140.45260	0.9928802	2684	Soil	18	6
## 24	C01_11	6.618126	220.66218	0.9954682	3076	Cotton	0	1
## 25	C02_14	6.627206	211.03921	0.9952615	3180	Cotton	0	2
## 26	C03_17	6.616958	216.06631	0.9953718	2938	Cotton	0	3
## 27	C04_20	6.626465	215.93901	0.9953691	3371	Cotton	0	4
## 28	C05_23	6.642822	211.08960	0.9952627	3435	Cotton	0	5
## 29	C06_26	6.679131	216.31351	0.9953771	3629	Cotton	0	6
## 30	C61_30	6.454741	170.03639	0.9941189	2767	Cotton	6	1
## 31	C62_33	6.484032	172.35279	0.9941979	3377	Cotton	6	2
## 32	C63_36	6.517958	173.41489	0.9942335	3804	Cotton	6	3
## 33	C64_39	6.476069	167.13138	0.9940167	3204	Cotton	6	4
## 34	C65_42	6.569722	197.01186	0.9949242	3250	Cotton	6	5
## 35	C66_45	6.482145	172.96394	0.9942184	3009	Cotton	6	6
## 36	C121_49	5.944568	71.55607	0.9860249	2779	Cotton	12	1
## 37	C122_52	6.187755	96.43939	0.9896308	3193	Cotton	12	2
## 38	C123_55	6.129460	81.26646	0.9876948	2859	Cotton	12	3
## 39	C124_58	6.028523	75.49726	0.9867545	2950	Cotton	12	4
## 40	C125_61	6.148179	98.94468	0.9898933	3018	Cotton	12	5
## 41	C126_64	6.347332	150.05708	0.9933359	2946	Cotton	12	6
## 42	C181_68	6.301392	132.36230	0.9924450	3266	Cotton	18	1
## 43	C182_72	6.000205	83.90929	0.9880824	2969	Cotton	18	2
## 44	C183_76	5.981284	82.44127	0.9878702	2636	Cotton	18	3
## 45	C184_80	5.578566	50.73174	0.9802885	2043	Cotton	18	4
## 46	C185_84	6.064655	87.82732	0.9886140	3113	Cotton	18	5
## 47	SB01_12	6.644864	216.86110	0.9953888	3203	Soybean	0	1
## 48	SB02_15	6.615662	211.32573	0.9952680	3055	Soybean	0	2
## 49	SB03_18	6.693987	230.45439	0.9956607	3595	Soybean	0	3
## 50	SB04_21	6.647502	234.80343	0.9957411	3253	Soybean	0	4
## 51	SB05_24	6.605749	198.57265	0.9949641	3187	Soybean	0	5
## 52	SB06_27	6.640696	215.26494	0.9953546	3190	Soybean	0	6
## 53	SB61_31	6.044229	89.13912	0.9887816	2371	Soybean	6	1
## 54	SB62_34	6.437589	154.21624	0.9935156	3248	Soybean	6	2
## 55	SB63_37	6.194632	83.11681	0.9879687	2976	Soybean	6	3
## 56	SB64_40	6.117393	87.20257	0.9885324	3006	Soybean	6	4
## 57	SB65_43	5.439798	29.48338	0.9660826	2809	Soybean	6	5
## 58	SB66_46	6.195816	108.22394	0.9907599	2680	Soybean	6	6

## 59	SB121_50	4.393341	12.39587	0.9193280	2508 Soybean	12	1
## 60	SB122_53	5.630929	52.97931	0.9811247	2403 Soybean	12	2
## 61	SB123_56	5.579523	48.59842	0.9794232	2752 Soybean	12	3
## 62	SB124_59	5.406651	34.08685	0.9706632	2946 Soybean	12	4
## 63	SB125_62	5.863941	63.33020	0.9842097	3165 Soybean	12	5
## 64	SB126_65	5.738025	57.88780	0.9827252	2705 Soybean	12	6
## 65	SB181_69	5.671024	57.37726	0.9825715	2642 Soybean	18	1
## 66	SB182_73	5.489406	43.16854	0.9768350	2773 Soybean	18	2
## 67	SB183_77	5.713960	60.47882	0.9834653	2454 Soybean	18	3
## 68	SB184_81	5.467076	44.06798	0.9773078	2365 Soybean	18	4
## 69	SB185_85	5.729473	55.95864	0.9821297	2789 Soybean	18	5
## 70	SB186_89	5.556356	54.34527	0.9815991	2050 Soybean	18	6
##	Water_Imbibed		even				
## 1		na	0.8171431				
## 2		na	0.8232216				
## 3		na	0.8046776				
## 4		na	0.8049774				
## 5		na	0.8192376				
## 6		na	0.8155427				
## 7		na	0.8125582				
## 8		na	0.8053387				
## 9		na	0.8057856				
## 10		na	0.8026420				
## 11		na	0.8134652				
## 12		na	0.8178151				
## 13		na	0.7825905				
## 14		na	0.8007927				
## 15		na	0.8027732				
## 16		na	0.8012745				
## 17		na	0.8019483				
## 18		na	0.8015438				
## 19		na	0.7934213				
## 20		na	0.7904154				
## 21		na	0.8168340				
## 22		na	0.8016534				
## 23		na	0.7965737				
## 24		0.0042	0.8240330				
## 25		0.0091	0.8217613				
## 26		0.0013	0.8286233				
## 27		0.0087	0.8157692				
## 28		0.0075	0.8158938				
## 29		0.0046	0.8148549				
## 30		0.058	0.8144250				
## 31		0.044	0.7980600				
## 32		0.0569	0.7906489				
## 33		0.0841	0.8022726				
## 34		0.0535	0.8124399				
## 35		0.0029	0.8093209				
## 36		0.0651	0.7496447				
## 37		0.0527	0.7668822				
## 38		0.0675	0.7702042				
## 39		0.0545	0.7545500				
## 40		0.0623	0.7673379				
## 41		0.0021	0.7945881				

```
## 42      0.0034 0.7787840
## 43      0.0632 0.7504026
## 44      0.0514 0.7593336
## 45      0.0577 0.7318864
## 46      0.0554 0.7539969
## 47      0.1664 0.8232153
## 48      0.0942 0.8244294
## 49      0.1248 0.8176063
## 50      0.115  0.8219646
## 51      0.0993 0.8188774
## 52      0.1005 0.8231136
## 53      0.2308 0.7777862
## 54      0.2603 0.7961603
## 55      0.2111 0.7744902
## 56      0.2808 0.7638754
## 57      0.2712 0.6850627
## 58      0.2887 0.7849191
## 59      0.2822 0.5612885
## 60      0.2557 0.7233538
## 61      0.2982 0.7044778
## 62      0.2489 0.6768294
## 63      0.2573 0.7275444
## 64      0.2285 0.7260697
## 65      0.2528 0.7197378
## 66      0.2706 0.6924349
## 67      0.3196 0.7320451
## 68      0.2437 0.7037462
## 69      0.2461 0.7221929
## 70      0.301  0.7286456
```

4 Question 4 - Using tidyverse language of functions and the pipe

```
library(tidyverse)
alpha_average <- alpha_even %>%
  group_by(Crop, Time_Point) %>%
  summarise(mean.even = mean(even, na.rm = TRUE), count = n(), sd.even = sd(even, na.rm = TRUE), se.even =
```

```
## 'summarise()' has grouped output by 'Crop'. You can override using the
## '.groups' argument.
```

```
alpha_average
```

```
## # A tibble: 12 x 6
## # Groups:   Crop [3]
##   Crop    Time_Point mean.even count sd.even se.even
##   <chr>      <int>     <dbl> <int>  <dbl>  <dbl>
## 1 Cotton         0      0.820     6 0.00556 0.00227
## 2 Cotton         6      0.805     6 0.00920 0.00376
## 3 Cotton        12      0.767     6 0.0157  0.00640
## 4 Cotton        18      0.755     5 0.0169  0.00755
```

```
## 5 Soil      0      0.814      6 0.00765 0.00312
## 6 Soil      6      0.810      6 0.00587 0.00240
## 7 Soil     12      0.798      6 0.00782 0.00319
## 8 Soil     18      0.800      5 0.0104  0.00465
## 9 Soybean   0      0.822      6 0.00270 0.00110
## 10 Soybean  6      0.764      6 0.0400  0.0163
## 11 Soybean 12      0.687      6 0.0643  0.0263
## 12 Soybean 18      0.716      6 0.0153  0.00626
```

5 Question 5 - Calculating difference between soybean column,soil column, and between cotton column and soil column

```
alpha_average2 <- alpha_average %>%
  select(Time_Point, Crop, mean.even) %>%
  pivot_wider(names_from = Crop, values_from = mean.even) %>%
  mutate(diff.cotton.even = Soil - Cotton, diff.soybean.even = Soil - Soybean)
alpha_average2
```

```
## # A tibble: 4 x 6
##   Time_Point Cotton  Soil Soybean diff.cotton.even diff.soybean.even
##       <int>   <dbl> <dbl>   <dbl>         <dbl>         <dbl>
## 1         0  0.820 0.814  0.822        -0.00602        -0.00740
## 2         6  0.805 0.810  0.764         0.00507         0.0459
## 3        12  0.767 0.798  0.687         0.0313         0.112
## 4        18  0.755 0.800  0.716         0.0449         0.0833
```

6 Question 6 - Connecting it to plots

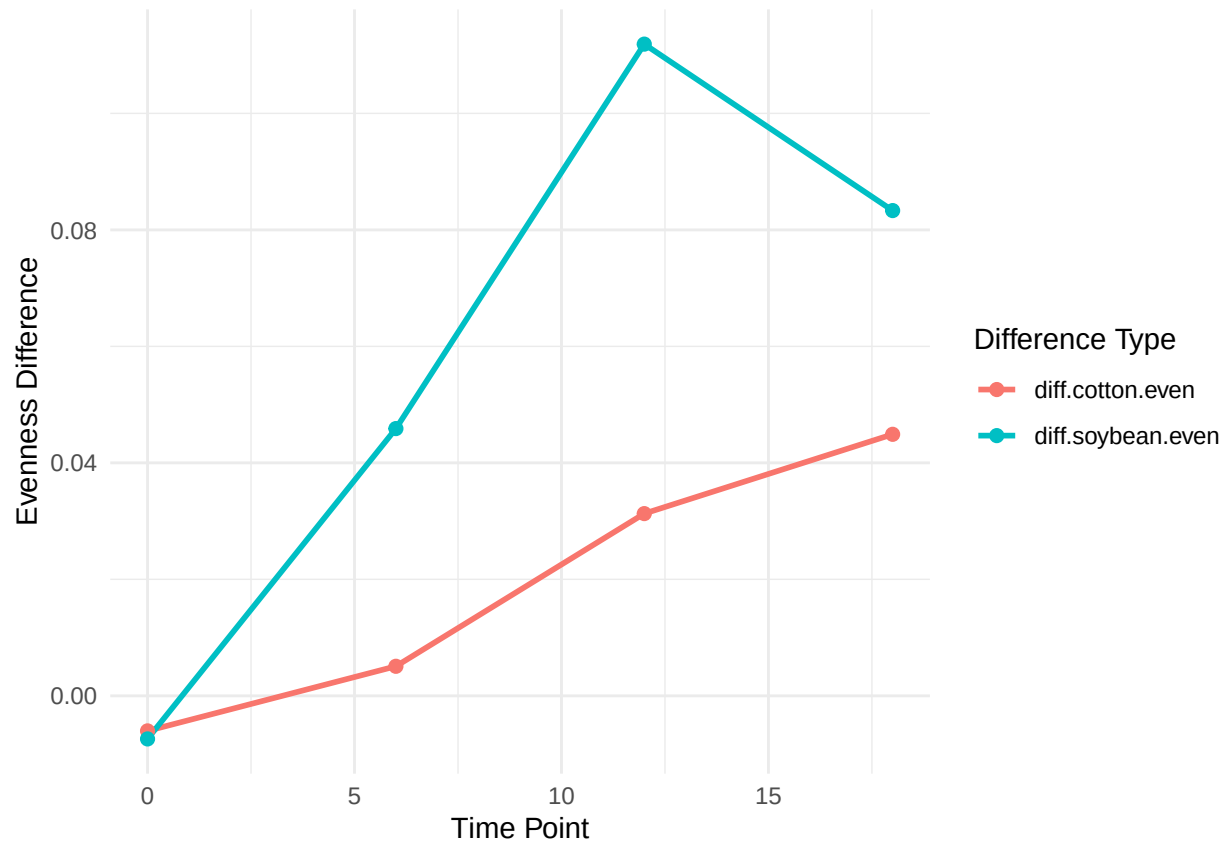
```
library(tidyr)
library(ggplot2)

alpha_plot_long <- alpha_average2 %>%
  select(Time_Point, diff.cotton.even, diff.soybean.even) %>%
  pivot_longer(cols = c(diff.cotton.even, diff.soybean.even), names_to = "diff", values_to = "values")

plot <- ggplot(alpha_plot_long, aes(x = Time_Point, y = values, color = diff)) +
  geom_line(size = 1) +
  geom_point(size = 2) +
  labs(x = "Time Point", y = "Evenness Difference", color = "Difference Type") +
  theme_minimal()
```

```
## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once per session.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.
```

```
print(plot)
```



7 Question 7 - My github assignment repository

<https://github.com/Hikmat-au/Assignments-repository>